

Section 1.3: The Least Squares Line

Finite Mathematics MTH 132 Spring 2014

Dr. James Ashe

Johnson C. Smith University

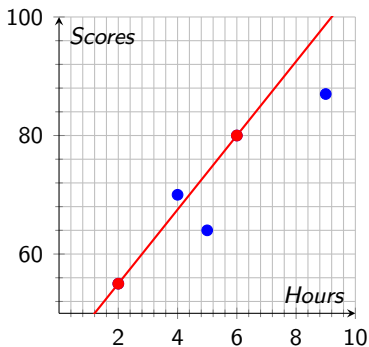
Often plotted data lies *close* to a line.

Often plotted data lies *close* to a line.

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

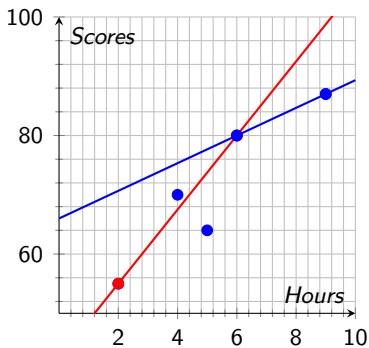
Often plotted data lies *close* to a line.

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87



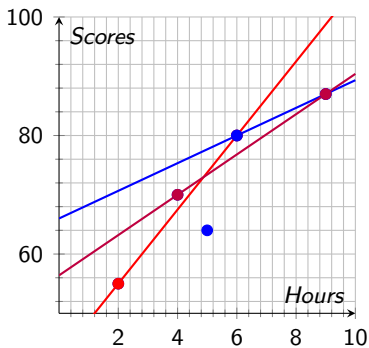
Often plotted data lies *close* to a line.

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87



Often plotted data lies *close* to a line.

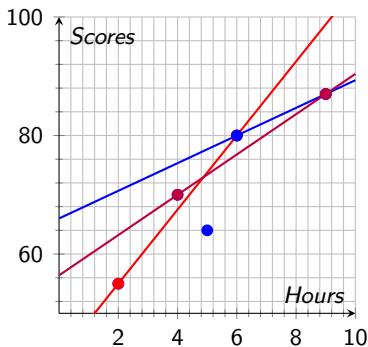
Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87



Often plotted data lies *close* to a line.

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

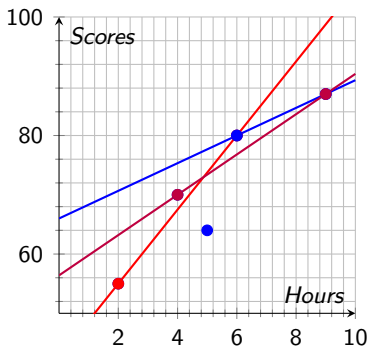
- We call this a *linear relation*.



Often plotted data lies *close* to a line.

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

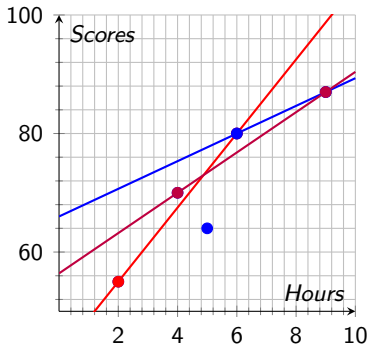
- We call this a *linear relation*.
- But which line?



Often plotted data lies *close* to a line.

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

- We call this a *linear relation*.
- But which line?
- We need a reasonable *approximation*.



The Least Squares Line (“best-fit line”)

The Least Squares Line (“best-fit line”)

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

The Least Squares Line (“best-fit line”)

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

Least Square Line *slope*

$$m = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Least Square Line *y-int*

$$b = \frac{\sum y - m(\sum x)}{n}$$

The Least Squares Line (“best-fit line”)

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

Least Square Line *slope*

$$m = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Least Square Line *y-int*

$$b = \frac{\sum y - m(\sum x)}{n}$$

Least Square Line

$$y = mx + b$$

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

Correlation Coefficient

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \sqrt{n(\sum y^2) - (\sum y)^2}}$$

Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

Equation of the least squares line

$$y = (4.54)x + 47.57$$

Correlation Coefficient

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \sqrt{n(\sum y^2) - (\sum y)^2}}$$

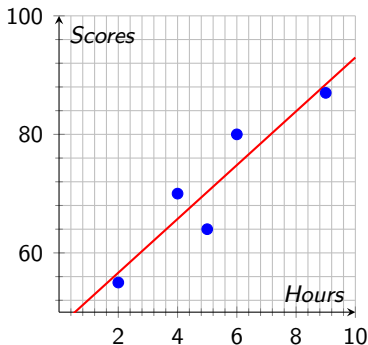
Hours Studying	Exam Scores
2	55
4	70
5	64
6	80
9	87

Equation of the least squares line

$$y = (4.54)x + 47.57$$

Correlation Coefficient

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \sqrt{n(\sum y^2) - (\sum y)^2}}$$



Problem 1.3.1. Find the equation of the least squares line.

The paired data below consist of the costs of advertising (in thousands) and the number of products sold (in thousands).

Cost(x)	9	2	3	4	2	5	9	10
Number(y)	85	52	55	68	67	86	83	73

Problem 1.3.1. Find the equation of the least squares line.

The paired data below consist of the costs of advertising (in thousands) and the number of products sold (in thousands).

Cost(x)	9	2	3	4	2	5	9	10
Number(y)	85	52	55	68	67	86	83	73

Problem 1.3.1. Find the equation of the least squares line.

The paired data below consist of the costs of advertising (in thousands) and the number of products sold (in thousands).

Cost(x)	9	2	3	4	2	5	9	10
Number(y)	85	52	55	68	67	86	83	73

$$y = 2.79x + 55.8$$

Problem 1.3.1. Find the equation of the least squares line.

The paired data below consist of the costs of advertising (in thousands) and the number of products sold (in thousands).

Cost(x)	9	2	3	4	2	5	9	10
Number(y)	85	52	55	68	67	86	83	73

$$y = 2.79x + 55.8$$

$$r \approx .708$$

